



Figure 1: Raspberry Pi and Python

that are available to perform various types of functionalities. Since this is outside the core language, you need to install these packages in your local development environment.

There are multiple Python packages that are made available in the Raspbian archives. And one of the recommended ways of installing these packages is via the standard `apt-get` commands as shown below:

```
sudo apt-get update
sudo apt-get install <python-package-name>
```

But not all packages are made available in the Raspbian archives and, in that case, it is advisable that you use the standard Python Pip package management system. For example, there is a Python requests package that makes it easy for Python applications to handle HTTP call functionality. We can install this package using the following command:

`$ pip3 install requests`

If you would like to see the current versions of the packages being managed by Pip, you can use the commands shown below. For instance, to check on the requests package that we have just installed in the previous step, use the following commands:

```
$ pip show requests
---
Name: requests
Version: 2.4.3
Location: /usr/lib/python3/dist-packages
Requires:
```

Writing a Python application

Python is a popular language for writing various utilities because of its simplicity and powerful packages. We shall write one such program to show you how you can write a simple Python application to get the current scores for all live cricket matches happening around the globe.

The steps for writing this program are as follows:

- 1) We will use the public API provided by cricapi.com.
- 2) The API endpoint at <http://cricapi.com/api/cricket> will give us a JSON response for all the current live matches, along with their scores.
- 3) We will then print out all the live matches happening,

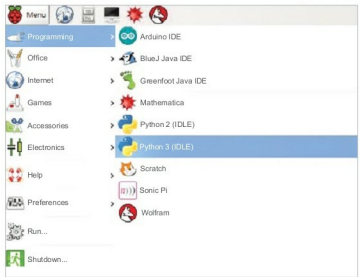


Figure 2: Starting Python 3 IDLE

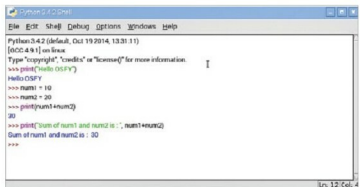


Figure 3: Python shell

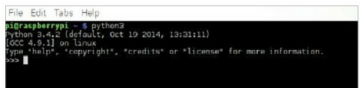


Figure 4: Python 3 shell in the terminal

along with their scores.

We can write this program in any editor provided on Raspberry Pi, like vi and nano. At the same time, we can also use the IDLE shell to write our program. We will do just that. Make sure that IDLE 3 is launched and from the shell's main menu, click on `File → New File (Ctrl + N)`.

This will open up an editor in which we can type out our program, as shown below:

```
import requests
import html

r = requests.get("http://cricapi.com/api/cricket")
if r.status_code == 200:
    currentMatches = r.json()["data"]
    for match in currentMatches:
        print(html.unescape(match["title"]))
else:
```